

M.E.R.LIN: Sistema de Assistência à Acessibilidade para Pessoas com Deficiência no Uso de Computadores

*M.E.R.LIN: Accessibility Assistance System for People
with Disabilities in Using Computers*

*M.E.R.LIN: Sistema de Asistencia a la Acessibilidad
para personas con Discapacidad en el Uso de
Computadoras*

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Resumo:

Este artigo contém a documentação escrita de um software que auxilia PCDs no uso de computadores e desktops. O objetivo é desenvolver um software assistivo que ofereça interfaces controláveis pelos movimentos faciais, voltado para pessoas com deficiências motoras, especificamente aquelas que não afetam os músculos mímicos, de modo a possibilitar que os indivíduos portadores dessas deficiências sejam devidamente inseridos no mercado de trabalho. A pesquisa realizada utiliza o estudo do método quantitativo, pois este permite verificar dados e resultados obtidos por meio de testes e observações das pessoas com deficiência motora no mercado de trabalho. Por meio dessa abordagem, foi possível analisar informações que possibilitaram medir o nível de acessibilidade e integração social de pessoas com deficiência motora ao mercado de trabalho nacional. Como resultado, espera-se que o software promova a inserção e a autonomia profissional do público-alvo, possibilitando uma melhora na qualidade de vida.

Abstract:

This article contains written documentation for software that assists people with disabilities in using computers and desktops. The goal is to develop assistive software that offers interfaces controllable by facial movements, aimed at people with motor disabilities, specifically those that do not affect the mimic muscles, to enable individuals with these disabilities to be adequately integrated into the job market. The research uses a quantitative method, as it allows for the verification of data and results obtained through tests and observations of people with motor disabilities in the job market. Through this approach, it was possible to analyze data that granted measuring the level of accessibility and social integration of people with motor disabilities in the national job market. As a result, the software is expected to promote the professional integration and autonomy of the target audience, thereby improving their quality of life.

Resumen:

Este artículo contiene documentación escrita sobre software que ayuda a personas con discapacidad a usar computadoras y ordenadores de escritorio. El objetivo es desarrollar software de asistencia que ofrezca interfaces controlables mediante movimientos faciales, dirigido a personas con discapacidades motoras, en particular aquellas que no afectan a los músculos mímicos, para facilitar su integración laboral. La investigación utiliza un método cuantitativo, ya que permite la verificación de datos y resultados obtenidos mediante pruebas y observaciones de personas con discapacidades motoras en el mercado laboral. A través de este enfoque, fue posible analizar información que permitió medir el nivel de accesibilidad e integración social de las personas con discapacidades motoras en el mercado laboral nacional. Como resultado, se espera que el software promueva la integración profesional y la autonomía del público objetivo, permitiendo una mejora en su calidad de vida.

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1. Introduction

M.E.R.LIN is a software capable of restoring independence in the use of desktop devices for people with physical disabilities, based on the concept of supported work, to develop assistive software that interacts directly with the computer's camera and operating system, allowing the execution of functions through a visual interface, with a wide range of actions to be performed through eye movement.

It can use eye movements to perform essential computer functions dynamically, facilitating the user's self-sufficiency and qualifying them to exercise their rights in society, such as the right to work and privacy.

Most companies are not concerned with increasing this group's employability in society; their primary concern is meeting the mandated percentage. This does not refer to hiring people with disabilities for tax exemptions. People with motor disabilities have minimal employment opportunities due to the lack of physical capabilities allowed to perform specific jobs, and workplaces are not regulated for people with motor disabilities, whether by choice or as a result of social programs. In short, the assistance system shows potential to increase the participation of people with motor dysfunctions in the workplace and to provide tools for inclusion in these spaces, both in terms of task efficiency and the social experience afforded by employment opportunities.

As Moreira et al. (2015 apud RODRIGUES; PEREIRA, 2021, p.9, our translation) state, "It is common for society, and in many cases the family itself, to reinforce the stigma that a person with a disability is incapable of performing daily or work activities independently and autonomously." Analyzing this societal paradigm, the project seeks to develop an application that confronts these precepts, delivering an inclusive and accessible solution for people with motor disabilities. The application is based on the concept of supported employment, aiming to become a model specialized system to assist people with motor dysfunction. "People with disabilities, often considered unfit for work, could perform work activities if they were provided with the necessary support" (Sousa, 2000, apud RODRIGUES; PEREIRA, 2021, p.15,our translation). Considering this, the project aims to be recognized as an assistant in the job market, promoting social interactions among this segmented part of society.

Exploring a recreational approach, the use of computer tools for socialization is also permitted. As presented by Lima (2013, apud RODRIGUES; PEREIRA, 2021), people with disabilities experience social experiences with greater intensity than others, resulting from a sense of social belonging and recognition of their work.

The following chapters of this article will address the principles behind the theoretical and technical foundation of this assistant, as well as the steps of its practical development. Among the fundamental technical concepts are Python (Banin, 2018), OpenCV (Castro, 2021), and Shell code (Negus, 2014).

Therefore, how can the use of facial assistive technology facilitate, include, and rehabilitate people with physical disabilities, permitting them to use eye movements to perform essential computer functions in a dynamic way that promotes user self-sufficiency and allows them to exercise their rights in society, such as the right to work and privacy?

2. Theoretical Framework

To provide a clear understanding of the concepts underlying the promotion of M.E.R.LIN, the following chapter outlines the theoretical and conceptual basis for its development.

2.1. Motor Disability in Brazil

Motor impairment represents a condition that significantly affects the quality of life and social participation of millions of Brazilians. According to data from the Brazilian Institute of Geography and Statistics (IBGE, 2023), approximately 8.9% of the national population lives with some form of disability,

corresponding to about 18.6 million people. This scenario underscores the need for public policies and affirmative actions to promote inclusion and accessibility. The physical limitations inherent in motor impairment frequently compromise the performance of daily and professional activities, underscoring the importance of developing and applying assistive technologies that promote the autonomy, self-sufficiency, and dignity of these individuals in social and economic contexts.

2.2. Python

Created by programmer Guido van Rossum during the 1990s for personal use, the programming language Python presents unique features in its design, as Banin (2018) points out. Among these unique features, Matthes (2016) cites Python's scalability, conciseness, and readability, making it flexible for developers. In accordance with Sweigart (2015), part of Python's flexibility stems from the large number of additional external resources available for building programs in this language.

2.3. Shell Script

As stated by Negus (2014), prior to the use of shortcuts and symbols, operating systems allowed communication between the user and the system's most primitive layer via a shell.

In his work, Neves (2010) demonstrates that a Shell, when used correctly, can be considered a programming language, as it allows multiple scripts, but is originally a simple command interpreter for the Operating System. While scripts, in line with Jargas's ideas (2008), are crude and executable codes that allow the performance of tasks, which can be simple or complex.

2.4. Computer Vision

As Barelli (2018) observes, computer vision is a great help in searching for, detecting, and processing information, analyzing it as human vision does and identifying characteristics.

As Antonello (2018) points out, computer vision goes beyond simply reading images; it can identify patterns, enabling the development of more complex systems, such as a fungus detector for plants, aimed at use in agriculture.

As Backes et al. (2016) state, computer vision can be used in more common applications not exclusively focused on the business sector, such as creating camera filters and performing image processing, such as noise reduction.

2.5. OpenCV

As Castro (2021) states, OpenCV is a library focused on developing computer vision applications, composed of modules that serve a wide variety of purposes in this area.

Marengoni (2010) emphasizes that the library, created by Intel, was designed to make real-time interaction between humans and machines more accessible. Similarly, as Barelli (2018) argues, the library's set of methods is highly effective for image manipulation, including the reading and interpretation of visual content.

2.6. MediaPipe

As claimed by Silva (2023), MediaPipe is a powerful tool for real-time detection and recognition, specialized in achieving high detection precision.

Schirmer (2023) emphasizes that the tool's range is approximately 4 meters and that, once the object or person is located, the system can continue to function efficiently up to 20 meters.

As maintained by (2025), the library provides simplified access to Machine Learning. Its portability and performance are factors for desktop and mobile development.

2.7. Database

In accordance with Silberschatz (2003), a database can be compared to an information warehouse, whose main objective is to improve the efficiency of locating, managing, and securing stored data.

As Louzada (2022) states, SQLite is a simple, practical, and intuitive database that does not require a client-server architecture.

Bannin (2018) emphasizes that, due to its ease of access and integration with Python, SQLite is an ideal database for Python applications. Its versatility allows it to be used across multiple platforms.

3. Method

The project adopts methodological definitions to underpin the procedures used in its studies of results, applications, and impacts. These are characterized as applied studies that aim to deliver a robust prototype to address the problem of digital accessibility for people with specific motor needs. The research approach is primarily quantitative, focusing on data related to user interaction. The statistical method is based on reducing social, political, and socioeconomic reports to quantitative data and on statistical manipulation that establishes and tests relationships and generalizations among phenomena, their meanings, or their natures. Through this approach, it was possible to analyze information that allowed measuring the level of accessibility and social integration of people with motor disabilities into the national labor market.

In the M.E.R.LIN project, the production of a bibliographic document allowed detailed research on the problem, the technologies used and implemented, and a greater understanding of the target audience's perspective, leading to greater proficiency in implementing an effective solution. Statistical methodology was used to verify the incidence of individuals in vulnerable situations in the job market regarding their personal needs related to computer or desktop use. This accurately captured how many people are in these situations, the type of disability, and the number of times companies hire people with disabilities (PWDs) due to legal requirements. It also showed that hiring people with motor disabilities is lower owing to a lack of company adaptability and the associated costs. To address these social shortcomings, a project was developed to reduce inequalities and effectively reach the target audience at a low cost.

4. Results and Discussion

As a result, the software allows the individual to be independent of peripherals such as a keyboard and mouse, requiring only an external or native camera on the device.

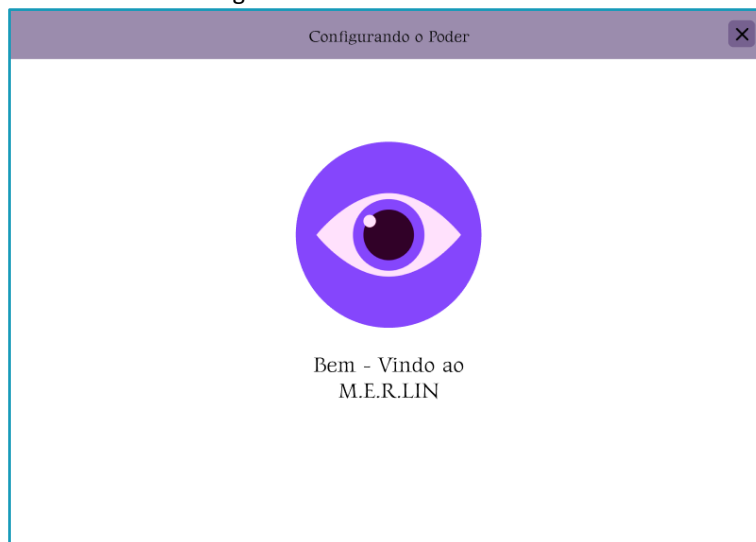
The project aims to use eye commands to control key computer actions, such as typing, searching, and opening and closing applications, in a customizable way for each user, requiring only an initial setup performed by a company technician or the user's guardian.

4.1. M.E.R.LIN

In this chapter, the resulting images of the work will be presented along with the context behind its development.

The first screen of the software is the startup screen, preceding the configuration and the first impression once the application is installed on the machine.

Figure 1 - Software home screen



Source: Author's own work, 2025.

After the initial screen, we have the configuration mode selection screen, where the user selects the configuration type: quick or custom.

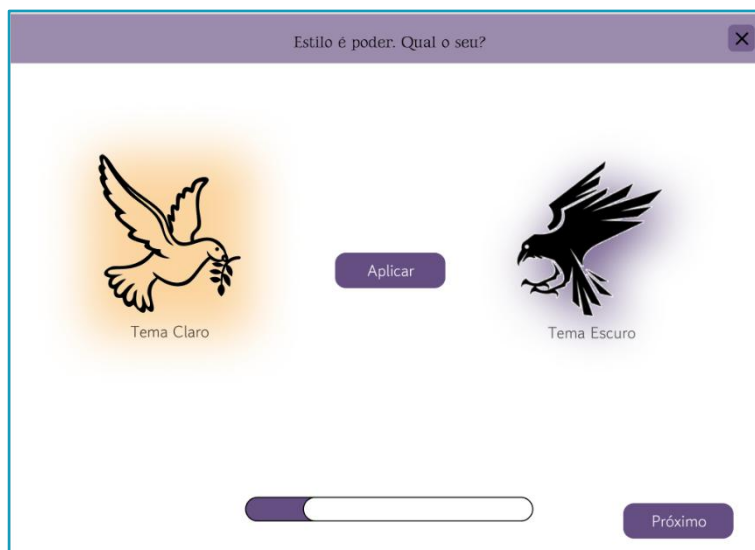
Figure 2 - Software configuration mode



Source: Author's own work, 2025.

If the technician or person in charge chose the quick setup option, the software configuration will end. However, if they choose the custom setup option, they will be directed to the theme selection page, where they can choose between dark and light.

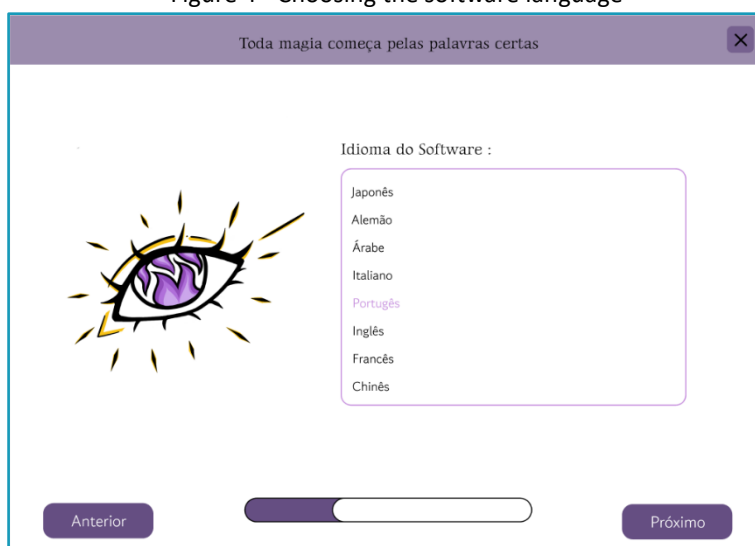
Figure 3 - Software theme



Source: Author's own work, 2025.

Next, you can choose the application's language.

Figure 4 - Choosing the software language



Source: Author's own work, 2025.

Next, after selecting the camera settings, the screen displays the Settings page, where you can define camera settings such as quality, FPS (frames per second), and the camera's alert light, which indicates whether it is on or off. On the settings screen, you can also reset the software language and view the terms of use.

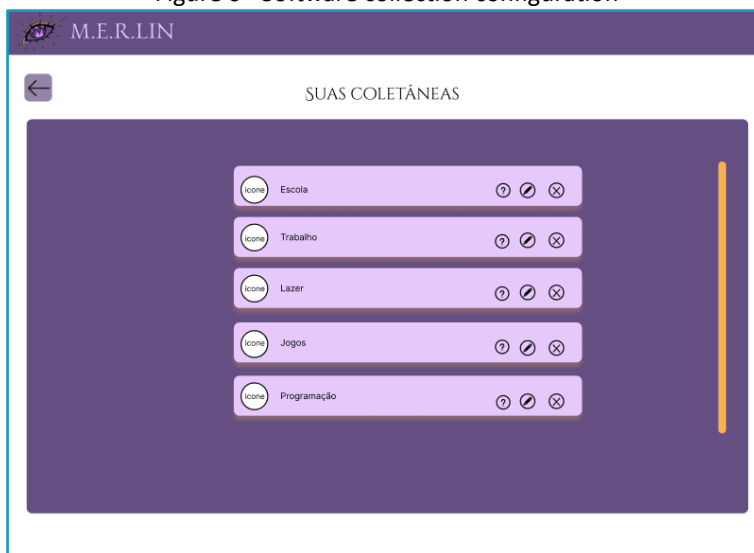
Figure 5 - Software camera settings



Source: Author's own work, 2025.

Finally, we will move on to defining collections, which are like application packages that can be opened with a specific command.

Figure 6 - Software collection configuration



Source: Author's own work, 2025.

If you choose to edit your collection, you will be directed to another page of collection commands; in the center, you will find five selectable application slots, allowing you to customize the application within the collection.

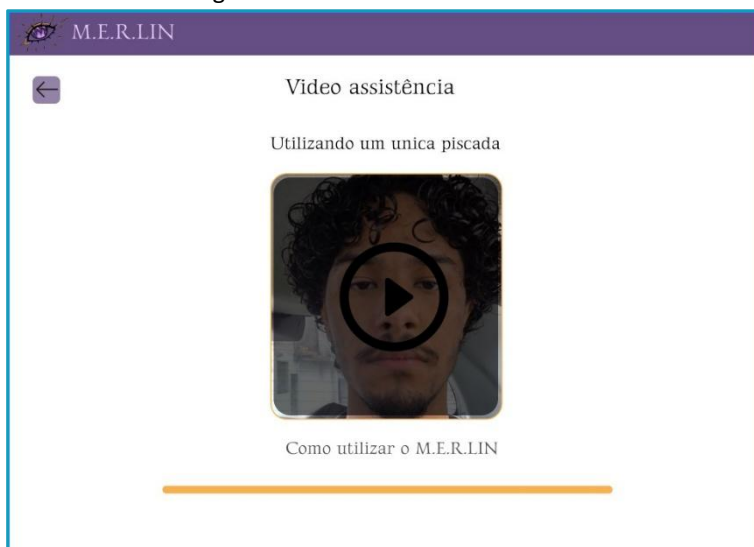
Figure 7 – Configuration of the software collection commands



Source: Author's own work, 2025.

If the option chosen is to view the command, the screen will display an assistance video showing how to perform the movement to execute that action.

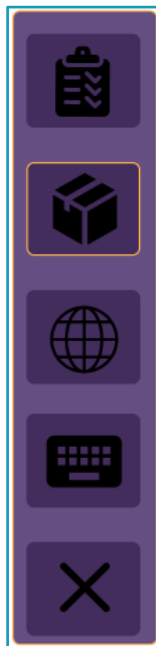
Figure 8 - Software assistance video



Source: Author's own work, 2025.

After the initial setup, the software is primarily guided by the Dock component, a panel that is always on-screen and shows the buttons that the software's eye commands can activate.

Figure 9 – Software Dock



Source: Author's own work, 2025.

5. Final Considerations

Based on quantitative research, a market needs and weaknesses analysis was conducted, with the expectation of delivering assistive software that could mitigate the aforementioned issues through facial recognition.

It was found during the completion of the M.E.R.LIN project that it has the potential to foster social interactions and job-market adaptation, since its initially proposed assistive software proposal was fully met.

Thus, it allows a focus on improving the project's usability and reach through integration with web browsers and keyboard suggestions. Furthermore, it is intended to expand M.E.R.LIN to support mobile devices, hence including new potential users.

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